

Semi-supervised segmentation of RNA 3D structures using density-based clustering

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ABSTRACT

A growing body of evidence shows that the biological activity of RNA molecules is not only due to their primary and secondary structures, but also to their spatial conformation. This is analogous to proteins, where investigating function, folding, or evolution often requires dividing the three-dimensional (3D) structure into subparts that can be studied individually. These independent substructures, known as protein “3D domains”, are geometrically defined as compact and spatially separate regions of the polypeptide chain. In RNA macromolecules, however, and to the best of our knowledge, no equivalent 3D-based concept has yet been formalized. We present RNA3DChast, an application of the Mean Shift clustering algorithm to the RNA 3D structure partitioning problem. For this work, a dedicated post-clustering procedure was developed to address the peculiarities of delimiting 3D domains in RNA conformations. Tuning and benchmarking RNA3DChast required us to create reference datasets of RNA 3D domain annotations and to devise a new scoring function—the Chain Segment Distance (CSD)—for assessing segmentation quality. Importantly, we show that the domain decompositions produced by RNA3DChast are consistent with those based on RNA biological function and evolution. Finally, the emerging interest in long non-coding RNAs (lncRNAs) and their likelihood of containing folded regions has motivated us to generate an additional reference dataset of lncRNA predicted conformations. The resulting delimitations of 3D domains by RNA3DChast illustrate the potential of our method for analyzing lncRNA 3D structures. Source code and datasets are freely available for download on the EvryRNA platform at: <https://evryrna.bio.univ-evry.fr/>.

1. Introduction

Analysis is defined as the separation of a whole into its constituent parts in order to gain a better understanding of it. In the context of structural biology, it consists of dividing macromolecules into simpler yet informative substructures, which can be studied individually. In the case of proteins, the spatial conformation of a polypeptide chain, which is integral to its biological function, can be described through different levels of analysis that include, hierarchically: (i) secondary structures, which are local folding patterns that emerge from interactions between nearby amino acids along the protein backbone, (ii) supersecondary structures, which are made of several adjacent secondary structure elements, and (iii) 3D domains—sometimes referred to as “structural domains”—which are regions of the protein that can fold, function or evolve independently. These latter subparts contribute to the dynamic nature of the protein, with “domain motion” being the process of 3D domain moving relative to each other, while keeping their individual

structures. Domains, rather than entire protein structures, are also responsible for mediating interactions between multi-domain proteins [1]. All these biological properties make the question of how to parse domains critical for understanding the molecular mechanisms that operate in the cell, and for potential applications, such as protein engineering or drug design. Importantly, protein 3D domains are the basic units used in many classification systems, such as the CATH [2], SCOP [3], and ECOD [4] databases.

In practice, 3D domains are central to the divide-and-conquer strategies employed for both *in vitro* determination and *in silico* prediction of protein structures. Experimental biologists often divide protein chains into autonomous folding units that are easier to purify and crystallize than the full-length macromolecule and yield higher-resolution crystallographic data. A single polypeptide chain can adopt multiple biologically active conformations due to inter-domain flexibility and motion. Therefore, rather than directly predicting the whole conformation, computational biology approaches predict the domain folds

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Proteins are the molecular machines of the cell

They carry out nearly all of its



Structural tasks

Building and maintaining the physical framework of the cell and its components



Functional tasks

The active processes that keep cells alive and responsive to their environment

Machines are built from different parts that each do a specific job

Proteins are often made up of distinct structural and functional units

→ The **domains**



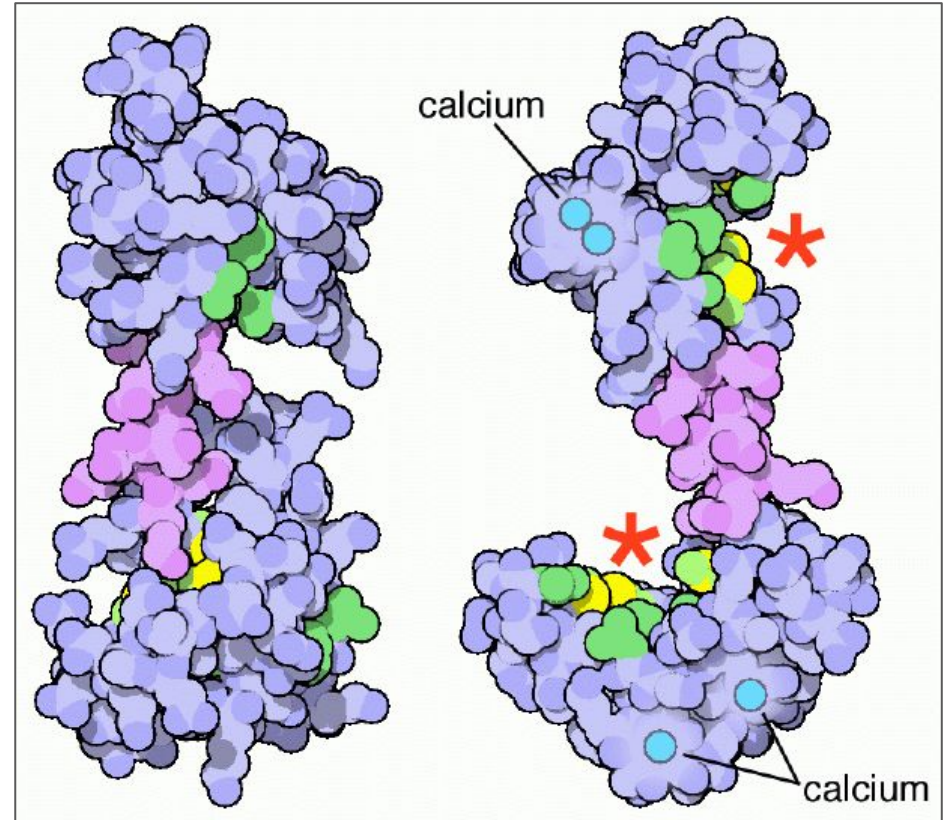
Protein domain: definition

Region of a protein that

- is compact,
- stable,
- can often fold independently,
- and perform a specific function.

Examples:

- binding DNA,
- attaching to a membrane,
- or catalyzing a chemical reaction.



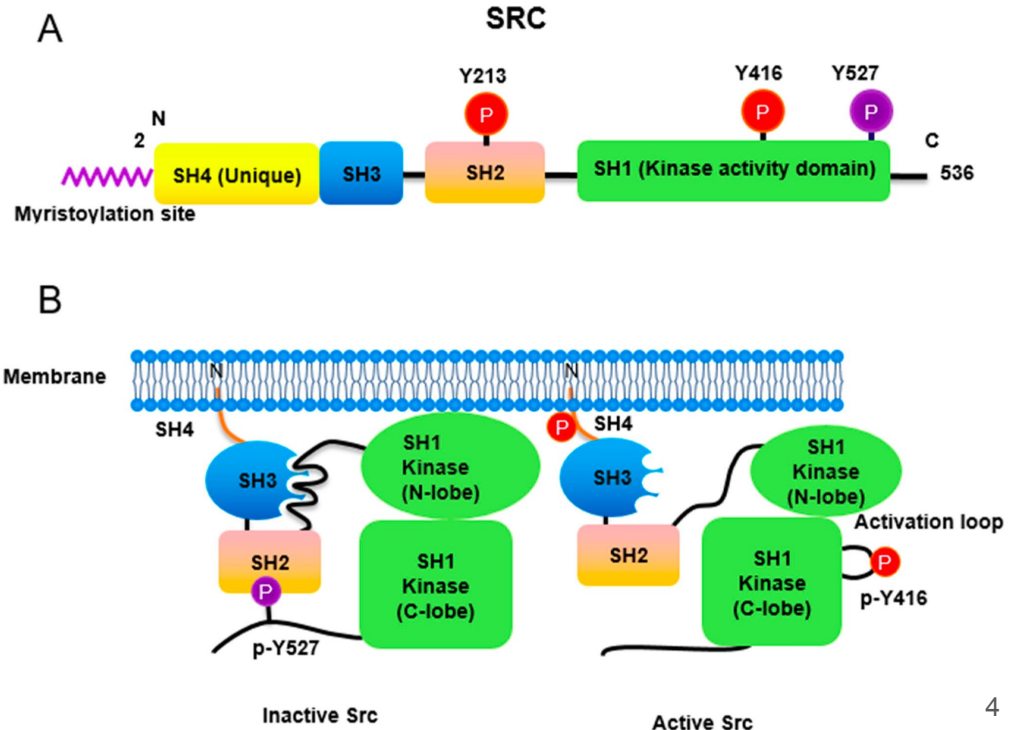
Calmodulin: inactive

active

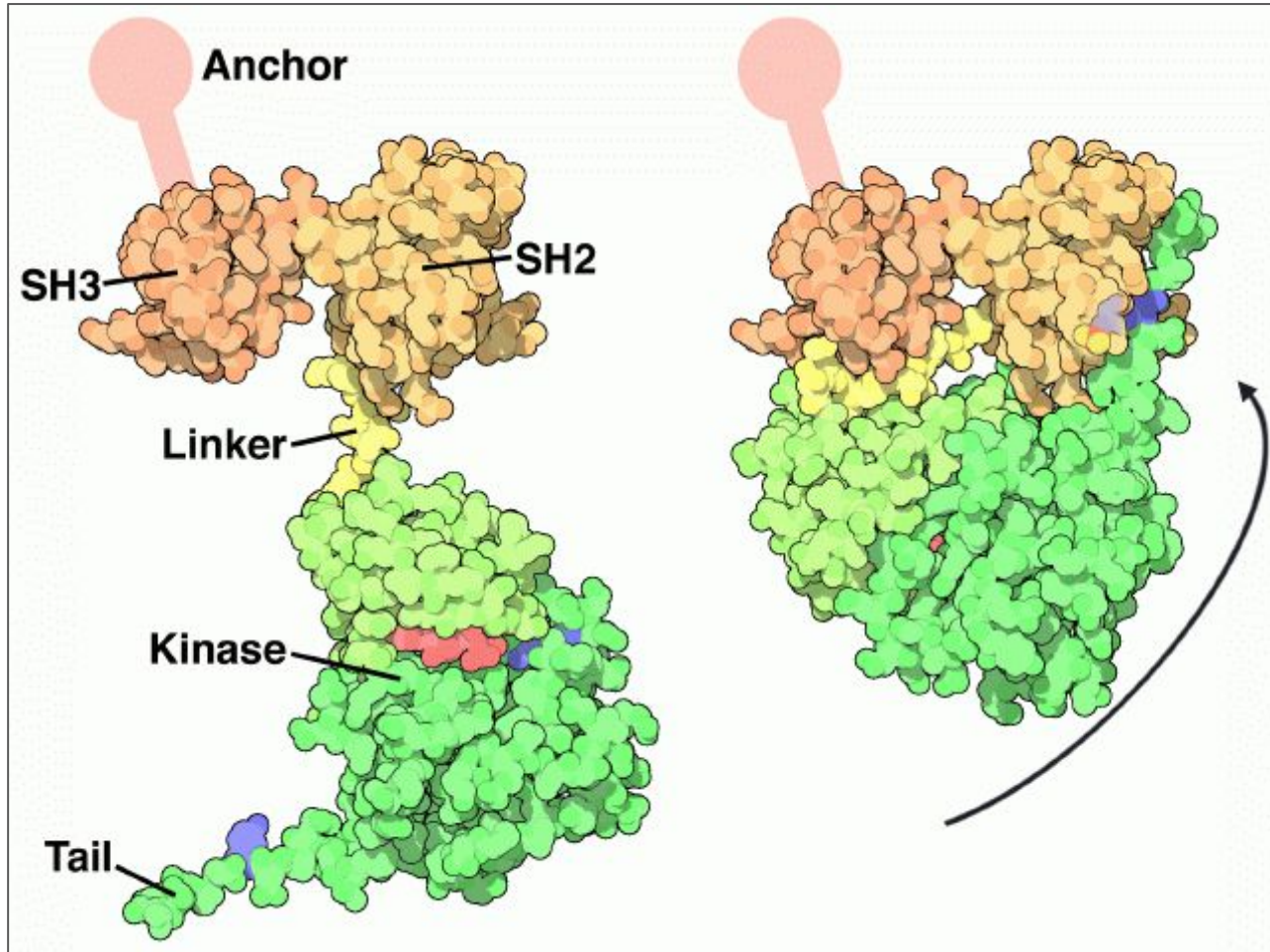
In the **Src tyrosine kinase**, there are several domains:

- SH3 domain: binds to proline-rich sequences,
- SH2 domain: binds to phosphorylated tyrosines,
- Kinase domain: catalyzes phosphorylation reactions.

Together, these domains allow Src to
sense signals and transmit them
through phosphorylation.



Each domain typically corresponds to a particular part of the protein's 3D structure

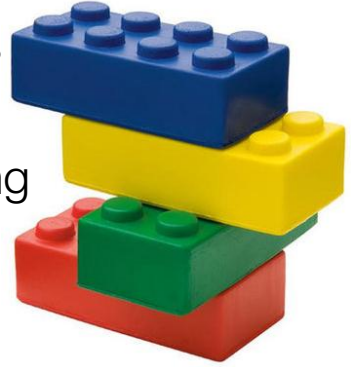


Modularity

Many proteins are built by combining domains in different arrangements

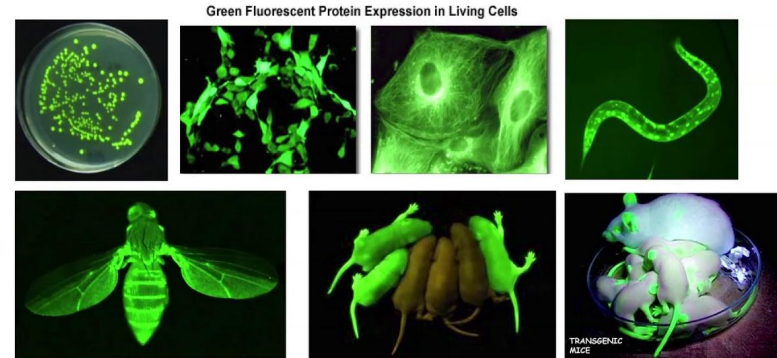
This allows to generate new proteins with novel functions by recombining existing domains

- Evolution
- Bioengineering



Examples:

- Fusion proteins such as GFP-tagged proteins, where a fluorescent domain (GFP) is fused to another protein to visualize its location in cells
- Multifunctional enzymes



How to delimit domains? (1/4)



Experimental and functional evidence

- **Limited proteolysis:** when treating a protein with mild proteases, flexible linkers are more easily cleaved, often revealing domain-sized fragments that remain stable.
- **Binding or enzymatic assays:** if a fragment retains activity or binding ability, it likely represents a functional domain.
Example: Cutting a DNA-binding protein and finding that one fragment still binds DNA → that fragment defines the DNA-binding domain.
- **Domain deletion or swapping** experiments: removing or fusing **suspected domains** can test whether they function independently.

How to delimit domains? (2/4)



Sequence-based bioinformatic methods

- **Homology searches:** comparing a protein's sequence against databases of known domains (e.g., Pfam, SMART, CDD).
- **Multiple sequence alignments:** conserved blocks often mark the boundaries of domains.
- **Hidden Markov Models (HMMs):** statistical models trained to recognize **conserved sequence motifs** and profiles typical of a domain family.
- **Secondary structure predictions:** **transitions** between α -helical and β -sheet patterns can indicate domain limits.

How to delimit domains? (3/4)

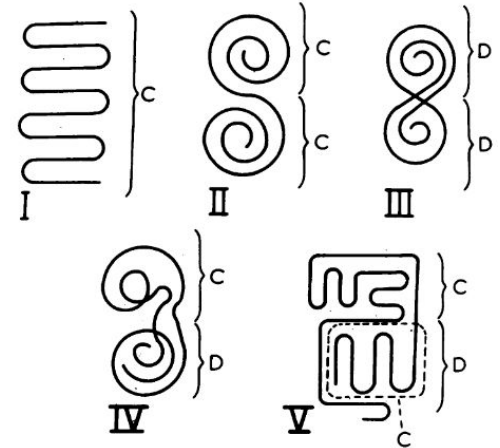


Structural criteria

Domains were **first defined** through protein 3D structures obtained by X-ray crystallography, NMR spectroscopy, and cryo-electron microscopy



A domain is a compact, independently folded region within a protein's 3D structure. Domains are often separated by flexible linker regions or loops.



How to delimit domains? (4/4)

Structural criteria

Algorithms for domain identification from 3D structures:

Algorithm	Principle	Key idea
PUU	Contact density	Compact units resist unfolding
DOMAK	K-means clustering	Geometric proximity
DETECTIVE	Contact matrix	Dense intra-domain contacts
DomainParser2	Graph partitioning	Minimizes inter-domain edges
DynDom	Motion analysis	Detects rigid-body domains
CATHEDRAL	Hybrid	Combines structure + sequence
SWORD	ML-based	Uses learned features

Domains in RNA

Rfam (short for RNA families)

The RNA equivalent of Pfam, the protein domain database

Rfam groups RNA sequences into families that share:

- sequence similarity
- secondary structure conservation
- and often, functional similarity



No 3D-based definition of RNA domains

Rfam HOME SEARCH PROJECTS FTP BLOG HELP

Motif: **Domain-V (RM00007)**
Description: *Splicing domain V*

0 sequences 0 queries 3 structures

Summary

Wikipedia annotation [Edit Wikipedia article](#)

The Rfam group coordinates the annotation of Rfam data in [Wikipedia](#). This motif is described by a Wikipedia entry entitled [Group II intron](#). [More...](#)

Group II introns are a large class of self-catalytic **ribozymes** and mobile genetic elements found within the **genes** of all three domains of life. Ribozyme activity (e.g., self-splicing) can usually occur under high-salt conditions *in vitro* for most group II introns. Certain highly reactive group II introns can undergo self-splicing under mild, near-physiologically relevant salt conditions. However, assistance from **proteins** is required for *in vivo* splicing. In contrast to **group I introns**, intron excision occurs in the absence of GTP and involves the formation of a **lariat**, with an A-residue branchpoint strongly resembling that found in lariats formed during splicing of nuclear pre-mRNA. It is hypothesized that pre-mRNA splicing (see **spliceosome**) may have evolved from group II introns, due to the similar catalytic mechanism as well as the structural similarity of the Group II Domain V substructure to the U6U2 extended **snRNA**. Finally, their ability to site-specifically insert into DNA sites has been exploited as a tool for biotechnology. For example, group II introns can be modified to make site-specific genome insertions and deliver cargo DNA such as reporter genes or *lox* sites

Structure and catalysis

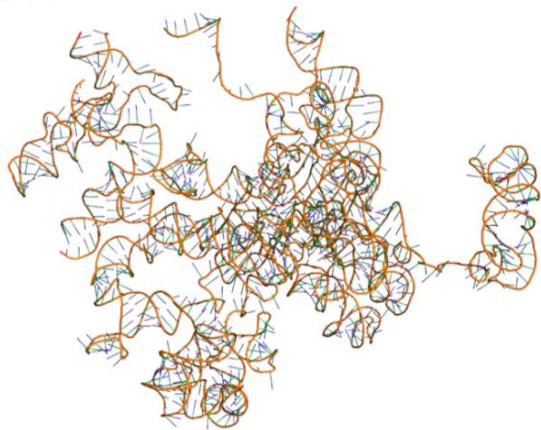
Group II catalytic intron, D5

full secondary structure of group II intron

Innovation: 3D domains in RNA structures

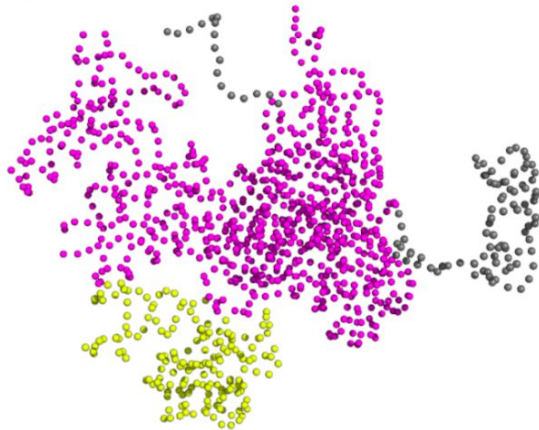
Example: LSub ribosomal RNA (rRNA) from *Trypanosoma brucei brucei*

A



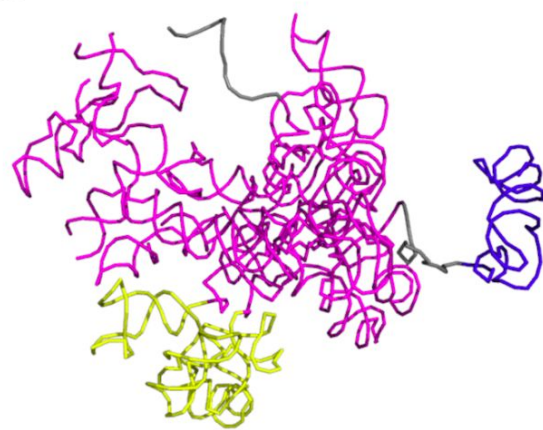
Cartoon representation

B



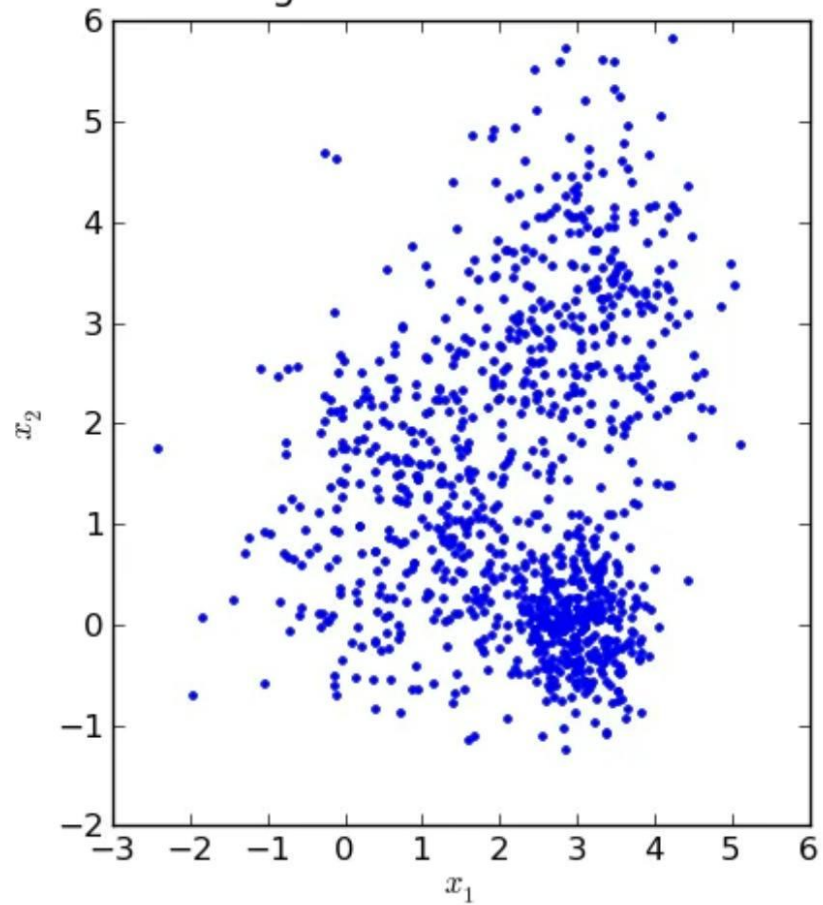
Clustering of atom 3D
coordinates
(single-atom representation)

C

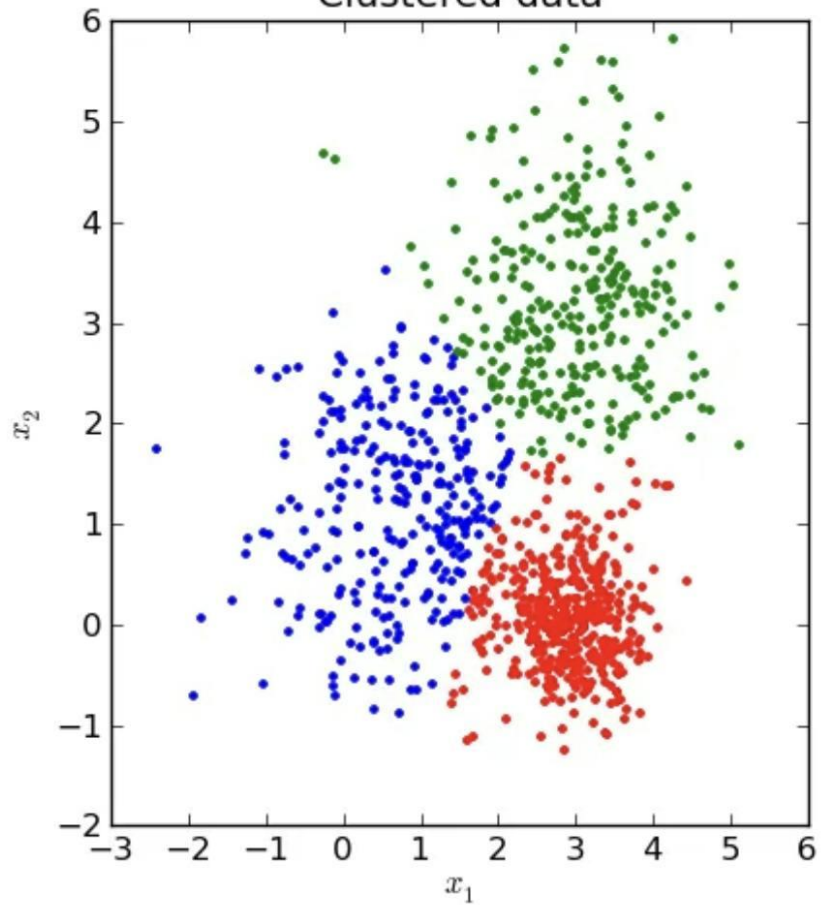


Post-processing

Original unclustered data

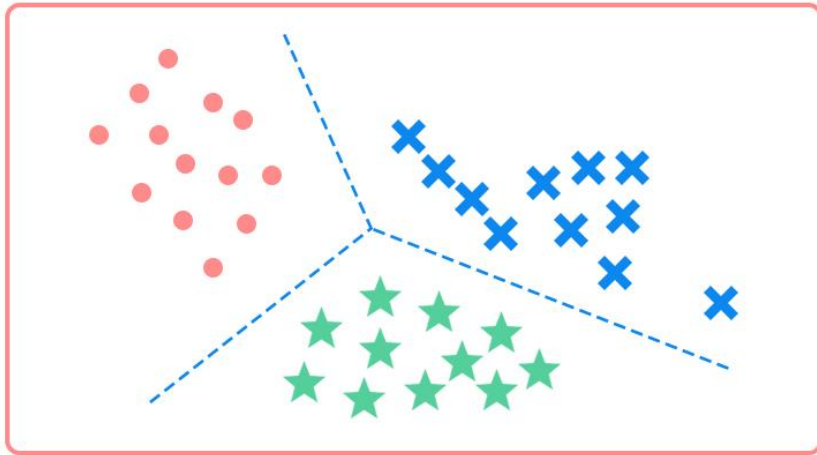


Clustered data



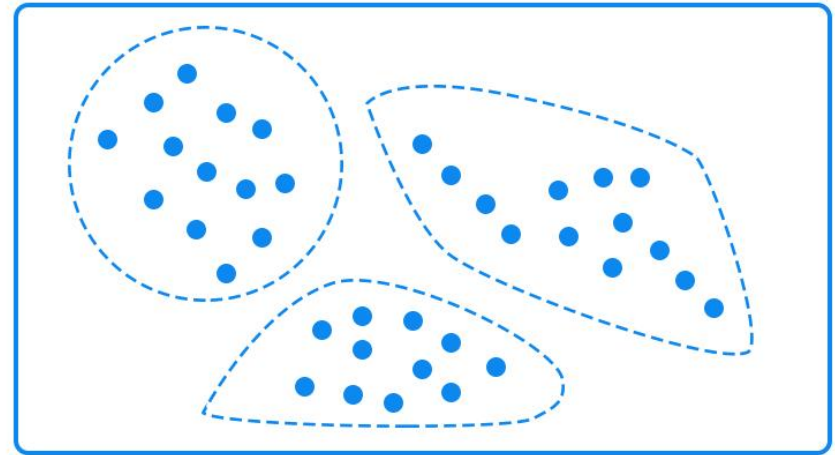
Supervised vs. Unsupervised Learning

Classification



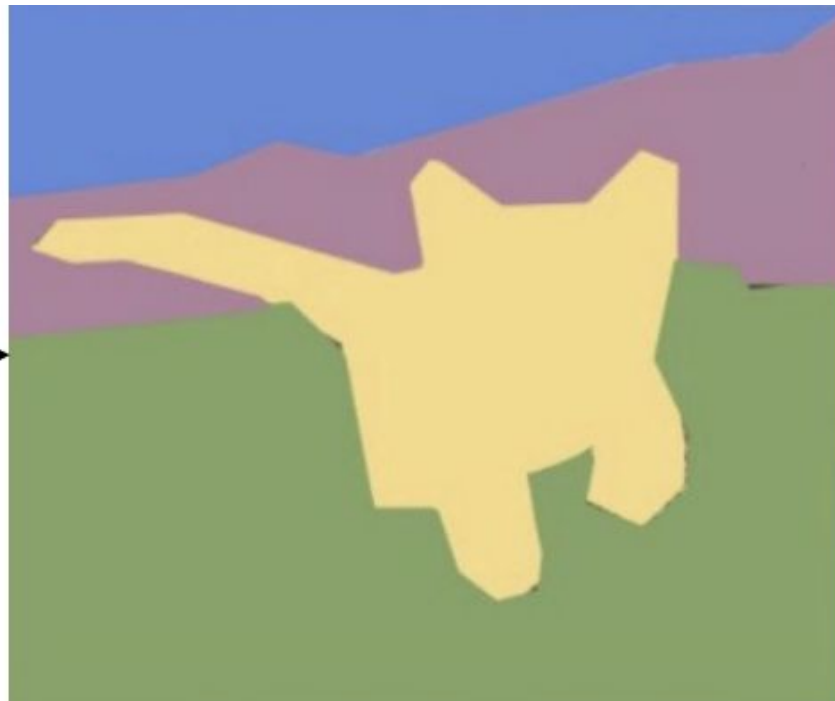
Supervised learning

Clustering

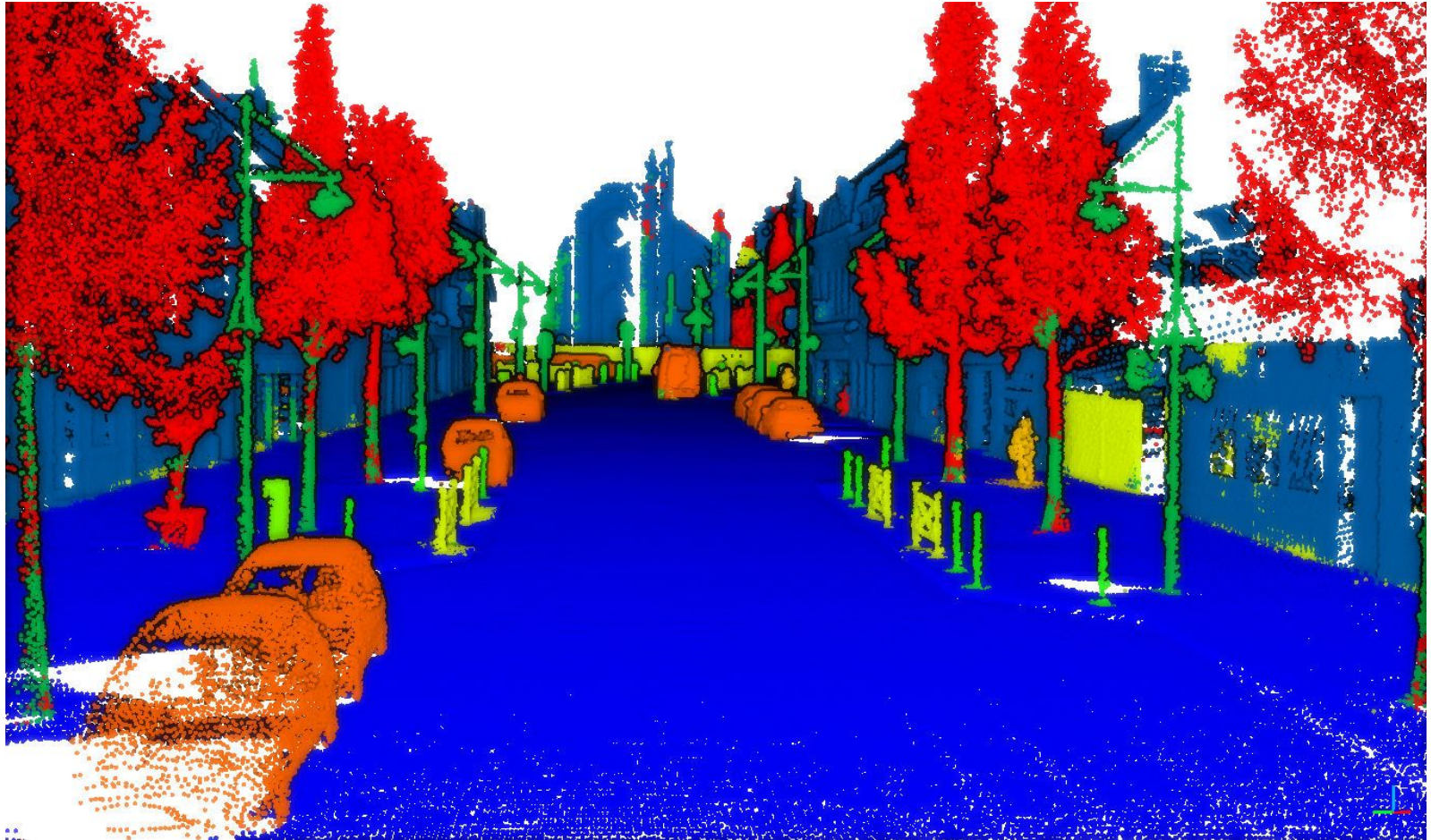


Unsupervised learning

Clustering for image segmentation

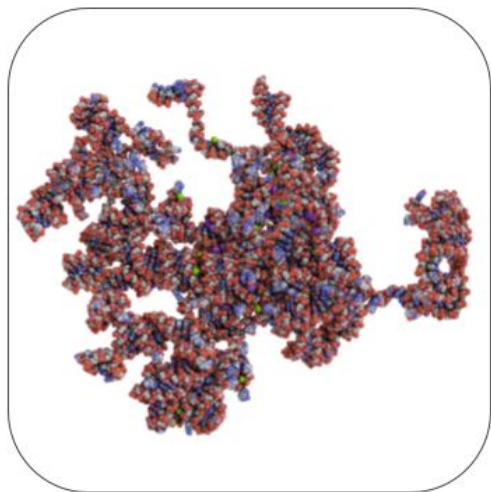


3D point cloud segmentation



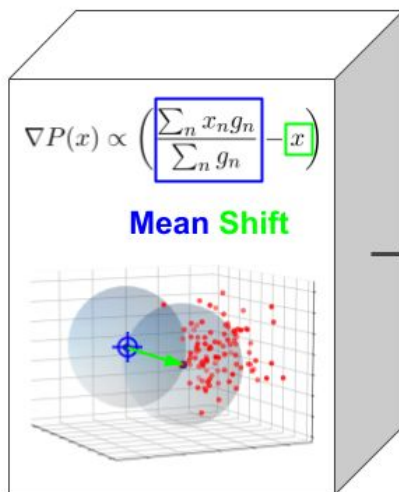
Density-based clustering for RNA 3D structure segmentation

Input

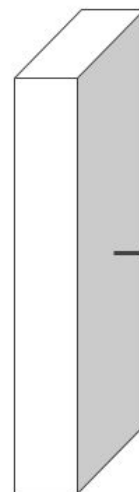


RNA 3D structure

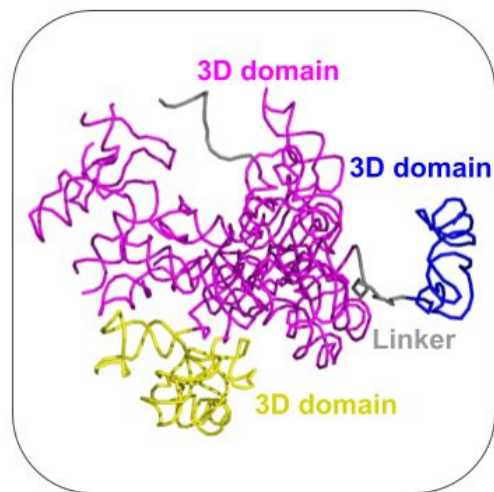
Clustering



Post-processing



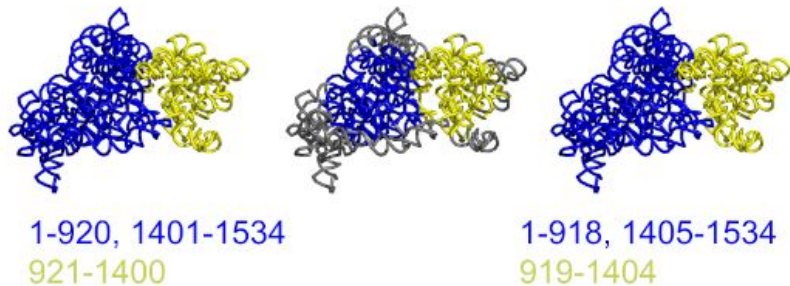
Output



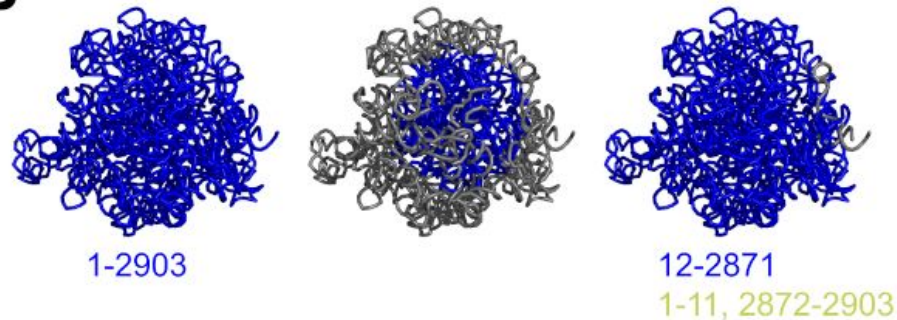
Segmentation

Density-based clustering for RNA 3D structure segmentation

A



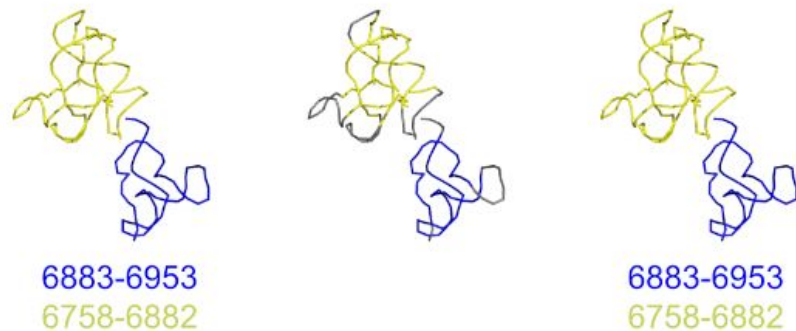
B



C



D

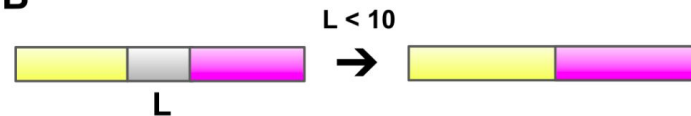


Post-clustering procedure: 8 rules

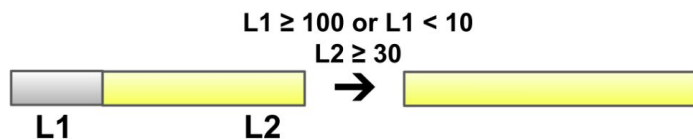
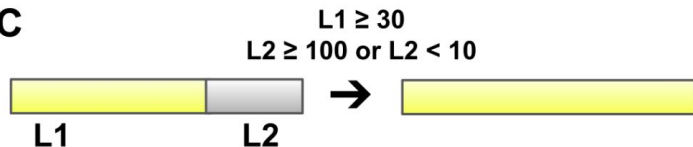
A



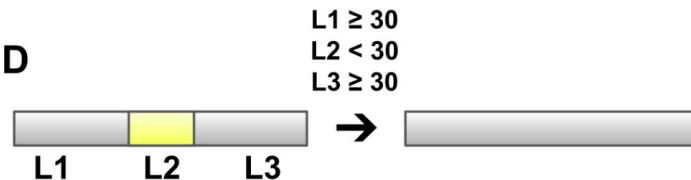
B



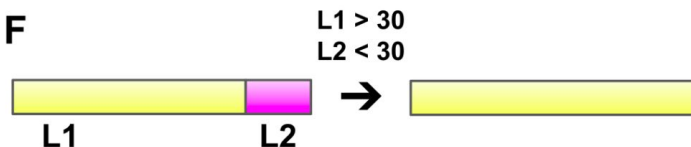
C



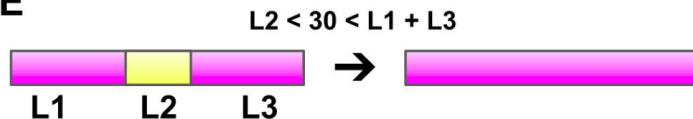
D



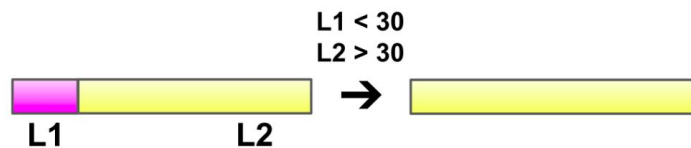
F



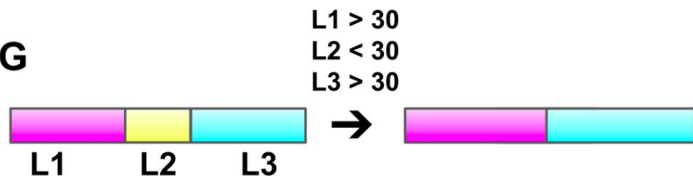
E



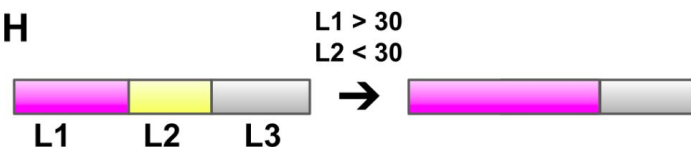
G



G

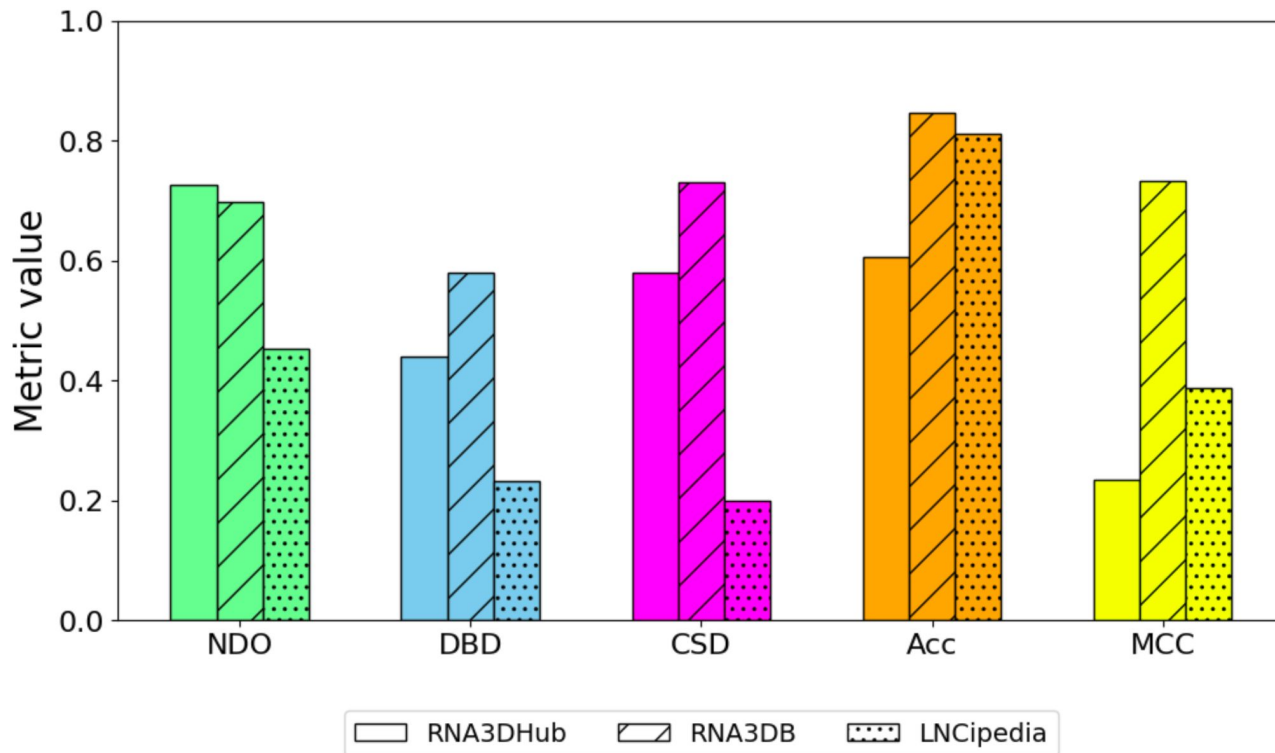


H



Performance: quantitative assessment

- 3 datasets
- 5 metrics (scoring functions)

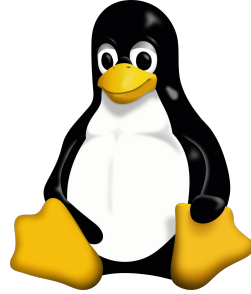


New scoring function: the Chain Segment Distance (CSD)

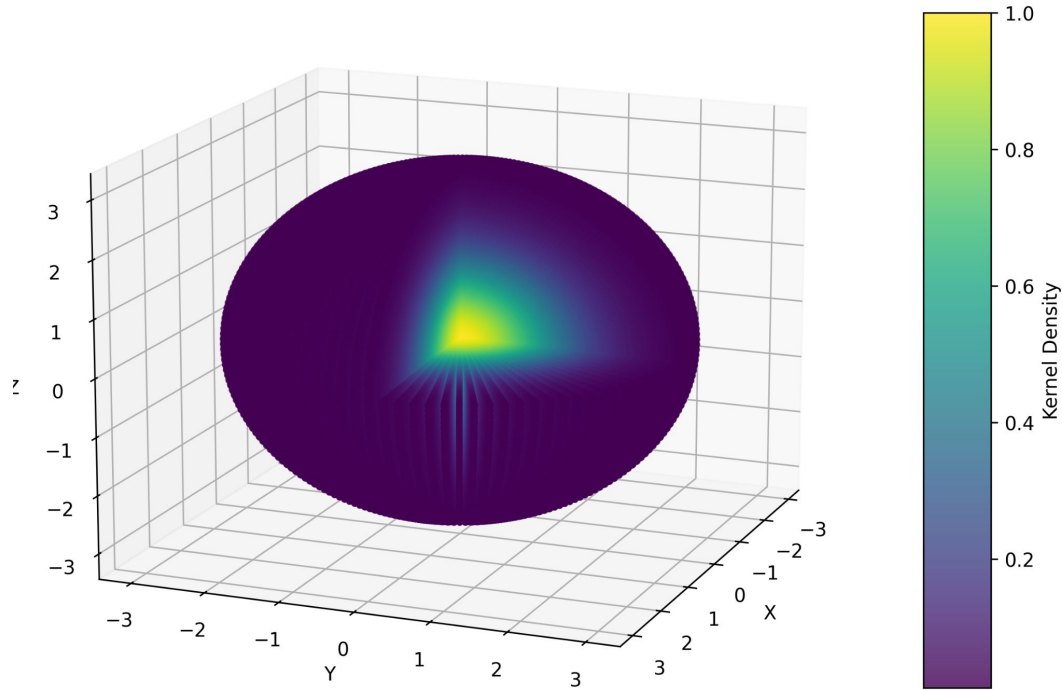
$$\text{CSD} = \begin{cases} \frac{\sum_{i=1}^m \max_{1 \leq j \leq n} S_{ij}}{T \times m}, & \text{if } m \geq n \\ \frac{\sum_{j=1}^n \max_{1 \leq i \leq m} S_{ij}}{T \times n}, & \text{otherwise} \end{cases}$$

$$S_{ij} = \frac{1}{2} \max \left\{ 2T - \left(d_{ij}^{5'} + d_{ij}^{3'} \right), 0 \right\}$$

Software and libraries



“4D normal distribution” used in Mean Shift clustering

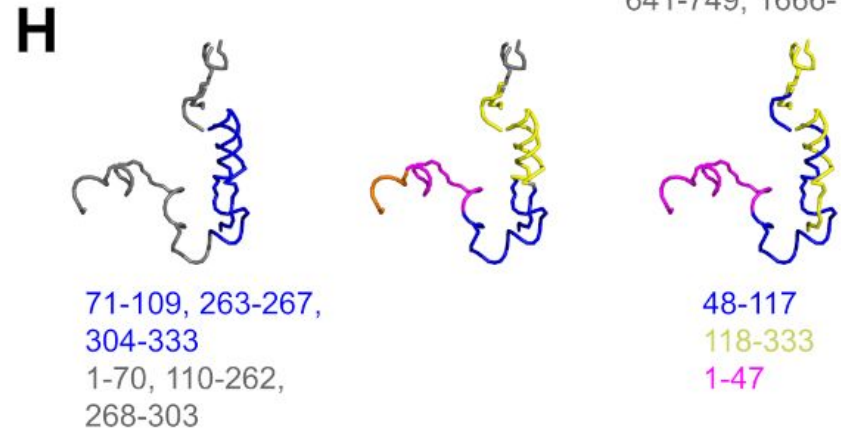
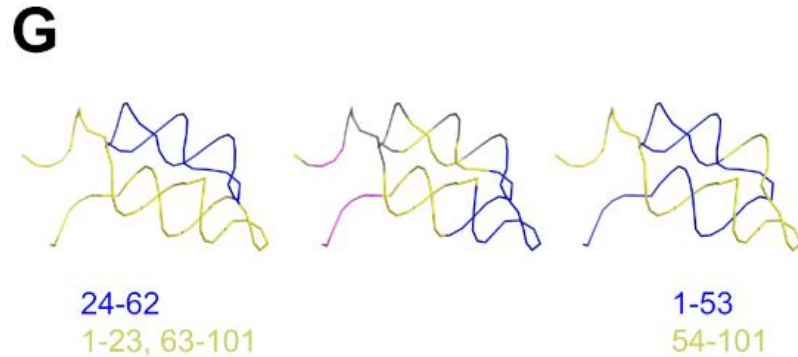
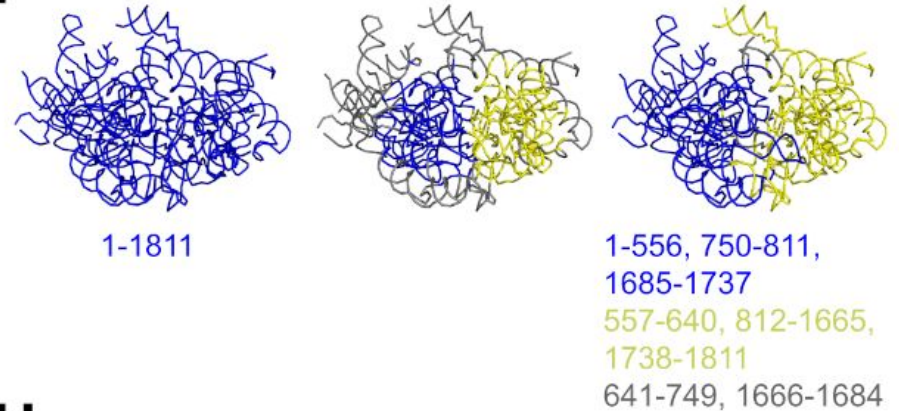


Still room for improvement...

E Under-segmentation



F Over-segmentation



Thank you for you attention!

